VITAMIN B12

Pernicious anaemia: lifelong IM

Hypokalaemia when starting

CHEMICAL NAME Cyanocobalamin ($C_{63}H_{88}CoN_{14}O_{14}P$); hydroxocobalamin preferred for injection in many countries

CLASS Water-soluble vitamin (cobalamin)

TRADE NAMES Cytamen (cyanocobalamin) · Neo-Cytamen (hydroxocobalamin) · Nascobal (nasal) · Cyanokit (hydroxocobalamin, cyanide antidote)

THERAPEUTIC USES / INDICATIONS

- Pernicious anaemia and B12-deficiency megaloblastic anaemia.
- B12 deficiency from gastrectomy, ileal resection, malabsorption, vegan diet, metformin, PPIs.
- Neurological complications of B12 deficiency.
- Cyanide poisoning (high-dose hydroxocobalamin).

PHARMACODYNAMICS

- Cofactor for **methionine synthase** (DNA synthesis — with folate) and **methylmalonyl-CoA mutase** (myelin synthesis) → corrects megaloblastic anaemia and neurological damage.

ADMINISTRATION

- **IM** (deep) or deep SC — preferred in malabsorption/pernicious anaemia.
- **PO** high-dose (1–2 mg daily) effective for dietary deficiency (and many malabsorption cases).
- Not IV for routine use (rapid renal loss).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain.
- Diarrhoea, itching.
- Pink/red urine (hydroxocobalamin).

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia in early treatment of severe anaemia (K^+ shifts into new cells).
- Anaphylaxis (rare).
- Polycythaemia (rare).

DOSAGES

- **Hydroxocobalamin (no neuro signs):** 1 mg IM three times weekly for 2 weeks, then 1 mg every 3 months for life.
- **With neurological involvement:** 1 mg IM alternate days until no further improvement, then every 2 months.
- **Cyanocobalamin IM:** 1 mg monthly (after loading).
- **Dietary deficiency (oral):** 50–150 mcg daily (or 1 mg daily).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to cobalamin or cobalt.
- Leber's optic atrophy (cyanocobalamin).

KEY INTERACTIONS

- **Metformin, PPIs, H2 blockers, colchicine, nitrous oxide:** ↓ B12 levels/absorption.
- **Chloramphenicol:** ↓ haematological response.

PHARMACOKINETICS

- Oral absorption requires intrinsic factor (terminal ileum); passive absorption ~1% of high oral doses.
- Stored in liver (years of reserve).
- Excreted in urine and bile (enterohepatic).

NURSING MANAGEMENT

- Check K^+ during first days of treating severe anaemia; monitor reticulocytes (peak 5–7 days).
- Neurological assessment (paraesthesia, gait, cognition).
- Schedule lifelong injections in pernicious anaemia.

PATIENT EDUCATION

- Pernicious anaemia needs injections for life.
- Tingling or numbness should improve slowly — report worsening.
- Vegans should take B12 supplements.

KEY POINTS

- B12; DNA synthesis and myelin.
- Treat B12 before folate.
- Pernicious anaemia: IM loading then 3-monthly for life.
- Watch K^+ early in severe anaemia.